SILVER LAYER

Data Cleaning, Standardization & Quality Control

SQL DATA WAREHOUSE PROJECT

SQL Server | Medallion Architecture

PURPOSE Document the layer workflow, key SQL patterns, validation checks, and implementation evidence.

1. Layer objective

The Silver layer converts raw Bronze data into consistent, reliable, and quality-checked datasets. Transformations should be repeatable, documented, and traceable to their source records.

2. Transformation workflow

Step	Transformation	Expected result
01	Read raw Bronze records	Source data available
02	Remove or flag duplicate business keys	Consistent record selection
03	Trim and standardize text values	Clean text attributes
04	Convert and normalize data types and dates	Consistent formats
05	Apply business rules and code mappings	Standardized values
06	Run quality checks and load Silver	Trusted Silver tables

3. Example: trim text values

TRIM removes leading and trailing spaces. Apply it to the relevant text columns in the actual transformation query.

```
SELECT
    cst_id,
    cst_key,
    TRIM(cst_firstname) AS cst_firstname,
    TRIM(cst_lastname)  AS cst_lastname
FROM bronze.crm_cust_info;
```

4. Example: choose the latest record

ROW_NUMBER can rank records within each business key. Confirm that the timestamp and tie-breaking rules match the source-system meaning.

```
SELECT *
FROM (
    SELECT *,
           ROW_NUMBER() OVER (
               PARTITION BY cst_id
               ORDER BY cst_create_date DESC
           ) AS row_num
    FROM bronze.crm_cust_info
) AS ranked
WHERE row_num = 1;
```

5. Data-quality checks

- Duplicate checks on business keys before and after deduplication.
- NULL checks for mandatory identifiers and required attributes.
- Whitespace and inconsistent casing checks on text columns.
- Date validity and date-range checks.
- Accepted-value checks for categorical columns.
- Row-count reconciliation with documented reasons for removals or exclusions.

6. Transformation inventory

Bronze source	Silver target	Rules applied	Validation result
[Actual table]	[Actual table]	[Actual rules]	[Actual result]
[Actual table]	[Actual table]	[Actual rules]	[Actual result]
[Actual table]	[Actual table]	[Actual rules]	[Actual result]

✓Completion target: Silver data follows documented cleaning rules, quality exceptions are explained, and downstream users can rely on standardized values.

 Verification note: examples illustrate common SQL techniques. Confirm exact transformation logic, target names, mappings, and execution results from the implemented Silver scripts.